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Late Jurassic/Early Cretaceous magnetic stratigraphy from the Sümeg section, Hungary

Emö Márton

Lorand Eötvös Geophysical Institute of Hungary, Budapest (Hungary)

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The pelagic limestones exposed at Sümeg appear to represent continuous deposition from Kimmeridgian through Berriasian. Detrital magnetite and haematite pigment are the carriers of remanence in the red and non-red limestones in the lower part, while magnetite becomes predominant in the upper half of the section. Thermal demagnetization has succesfully removed overprint magnetizations, and a well-defined magnetic stratigraphy has been obtained. The Late Jurassic/Early Cretaceous mixed polarity interval is correlated with the sequence of geomagnetic reversals derived from oceanic magnetic anomalies.

1. Introduction

The sequence of reversals of the geomagnetic field for the Late Jurassic/Early Cretaceous has been deduced from the lineated magnetic anomaly systems in oceans [1]. The dating of this reversal pattern is far from exact, for it is based on linear interpolation between two points, one in the Hauterivian and one in the Oxfordian. Moreover, the dating of those two points is debated [2]. The geomagnetic polarity sequence for the same period is not very well known from land sections although detailed magnetic stratigraphy studies have lately been presented for the Lower Cretaceous from the Cismon section [3] and from Umbria [4], both in Italy, for the Upper Jurassic from the Morrison Formation, U.S.A. [5] and from the Valdorbia section, Italy (J.E.T. Channell et al., personal communication). The main reason for the scarcity of palaeomagnetic information for the Late Jurassic/Early Cretaceous is that very weakly magnetized rocks were formed during this period. The development of extremely sensitive magnetometers now permits the investigation of these rocks but the accumulation of data has been slow. One reason for this is that the demagnetization response of the samples is often more positive to the time-consuming thermal treatment than to the faster AF technique. The purpose of this paper is to establish a land-based reversal stratigraphy for a poorly known part of the Mesozoic polarity time scale, e.g. Upper Kimmeridgian/Berriasian.

2. Geology and sampling

The sampled section is in the Transdanubian Central Mountains near the town of Sümeg (Fig. 1). The Oxfordian through Valanginian sequence consists of strongly weathered red shales in the Oxfordian and Lower Kimmeridgian, red limestones in the Upper Kimmeridgian and Lower Tithonian, and white to light grey biancone limestones and marls with chert nodules in the Tithonian through Valanginian [6]. The sedimentation is supposed to have been continuous with a very high mean sedimentation rate, around 15 m/Myr.
range from 2 to 20 cm. The beds are near vertical in a slightly overturned position. The variation of

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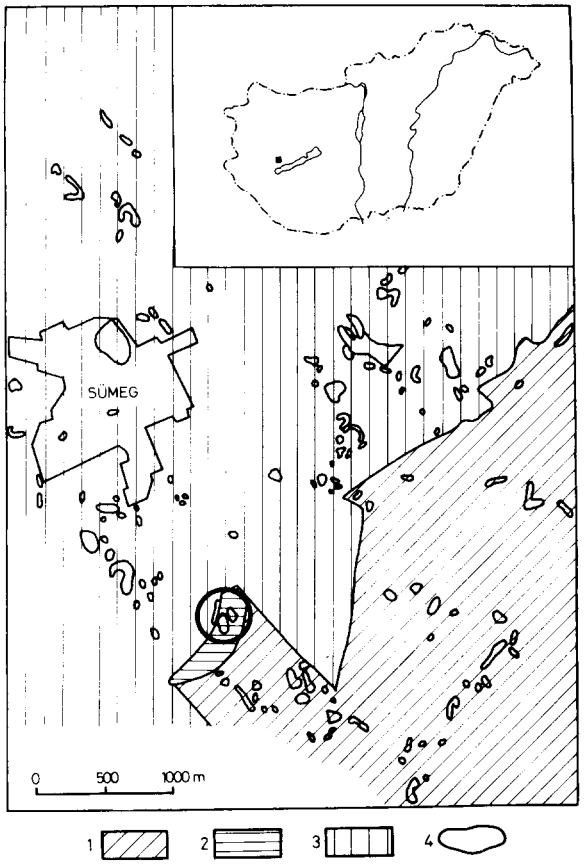


Fig. 1. Location map for the Sümeg section (circled). *1* = Triassic sediments; *2* = Jurassic sediments; *3* = Cretaceous sediments; *4* = outcrop.

the bedding attitude is generally slight within the 140-m-thick sampled interval of Upper Kimmeridgian through Lower Berriasian age.

Two disturbed areas do occur, where the azimuth is unchanged while the dips are more gentle, suggesting crenulations of the less competent beds within the more rigid surroundings.

158 samples were drilled at intervals of about 1 m in the field.

Samples younger than Berriasian were not collected because the beds are intensively distorted.

3. Magnetic properties

The isothermal remanence (IRM) acquisition curves give an estimate of the coercivity spectrum of the magnetic minerals present. Low-coercivity components (saturation in fields of 100–200 mT) are dominant in all samples investigated, but high-coercivity components contribute to the IRM of the samples at the base of the section, too (Fig. 2, 15.00 m).

The natural remanent magnetization (NRM) of several pilot specimens were demagnetized stepwise in a TSD-1 Schonstedt oven. In order to monitor mineralogical changes, susceptibility was measured (KLY Kappa Bridge) in many samples after each heating step, or in the natural state and after the last heating step. All remanence measurements were made using JR-4 Spinner (noise level 4×10^{-6} A/m). Magnetizations close to the noise level were measured repeatedly (each component 4 to 10 times) to check internal consistency.

As seen from the decay of NRM during progressive thermal demagnetization (Fig. 3), magnetite is one of the carriers of NRM (the low-coercivity mineral of the IRM curves). The high-coercivity phase of the IRM curves (pigment haematite) plays an important role in NRM too: the large amounts of directional movements above 300°C observed in a few pilot specimens (Fig. 4, 15.20, 19.20, 47.80 m) can be attributed to the removal of the overprint magnetization residing in

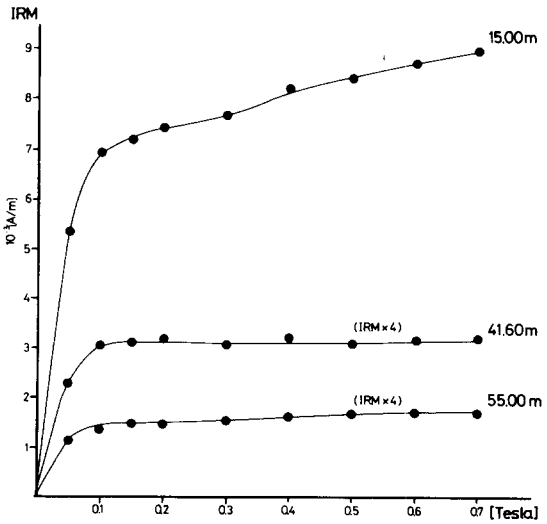


Fig. 2. Isothermal remanence (IRM) acquisition curves for three samples from the Sümeg section.

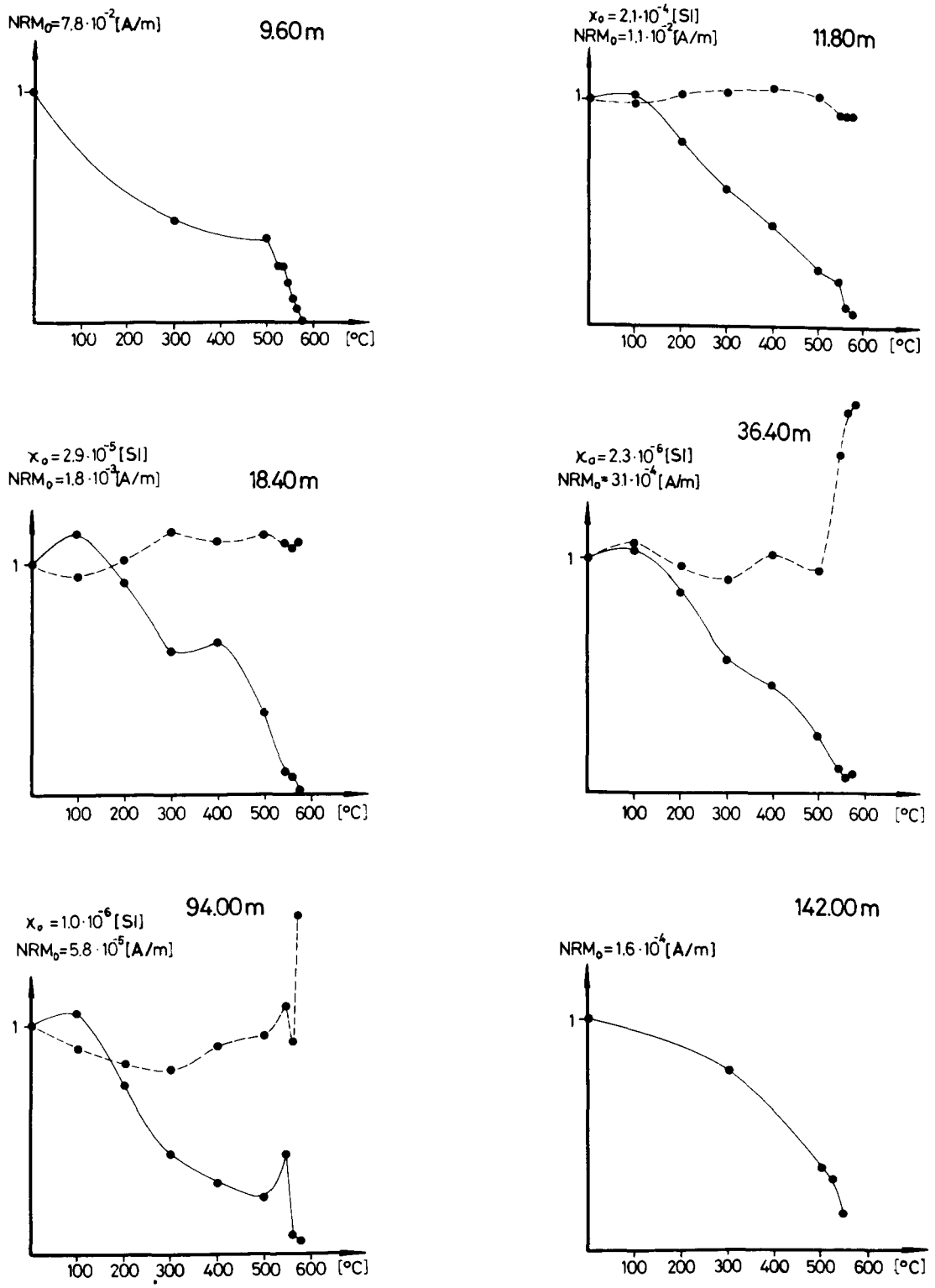


Fig. 3. Decay of the NRM of six samples during progressive thermal demagnetization (solid line) and change in susceptibility (broken line).

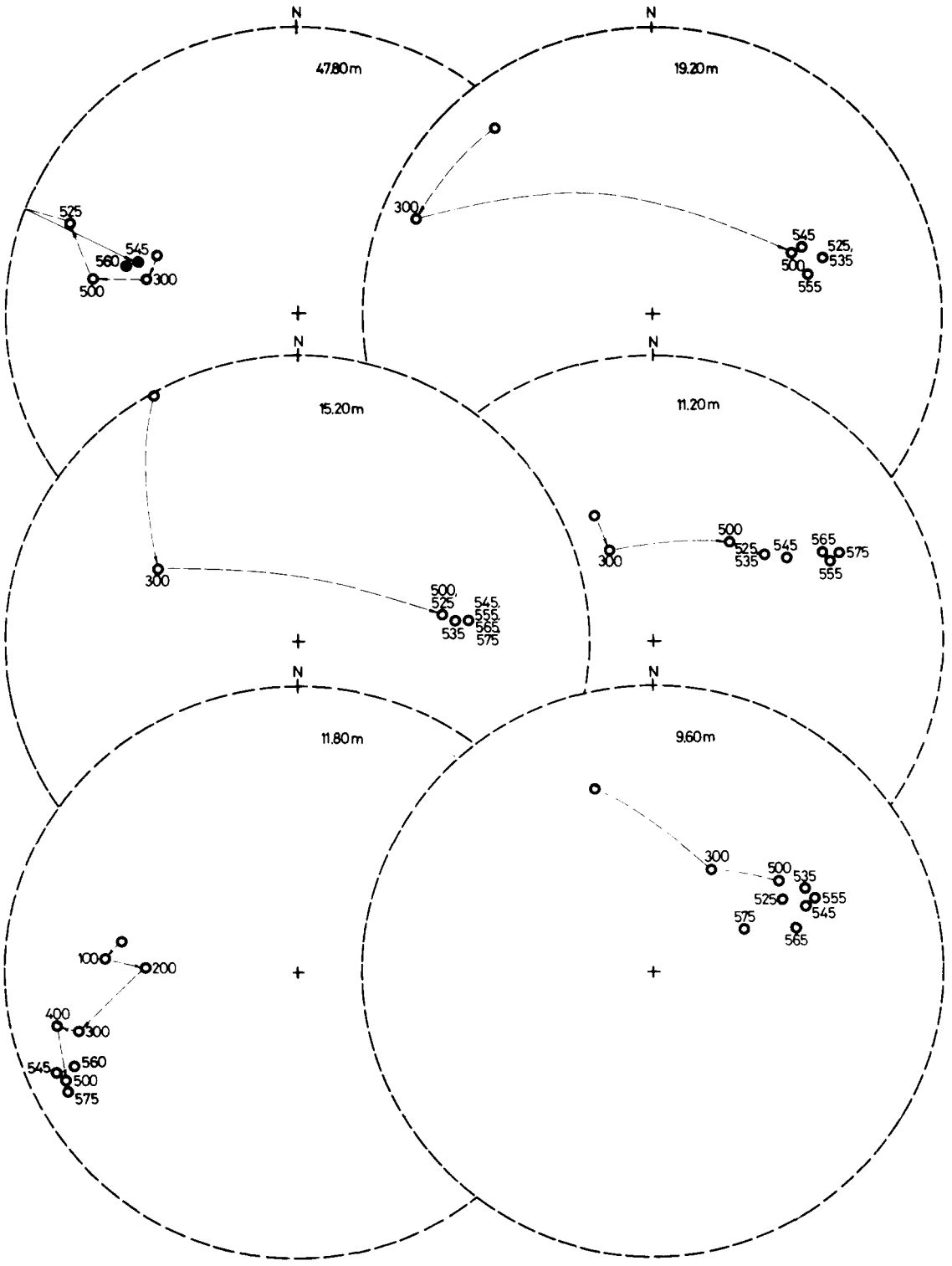


Fig. 4. Directional change in NRM of a few specimens on progressive thermal demagnetization. Stereographic plot. The directions are referred to the horizontal plane at the time of formation. Full circles: vectors pointing downward; open circles: vectors pointing upward.

haematite pigments. There is no indication of hydrous iron oxides.

The NRM reaches a stable end point at or above 500°C, but definitely below the Curie point of the magnetite (Fig. 4). In many samples, both red- and white-coloured (red samples below 20.00 m, greyish white above), a well-defined magnetization survives heating to 575°C. The polarities of the remanence, probably associated with dominant magnetite (generally between 500 and 575°C, Fig. 3) and high-blocking haematite ($T >$ 575°C), are identical (Fig. 4). AF demagnetization gives dubious results. It is true that in many cases

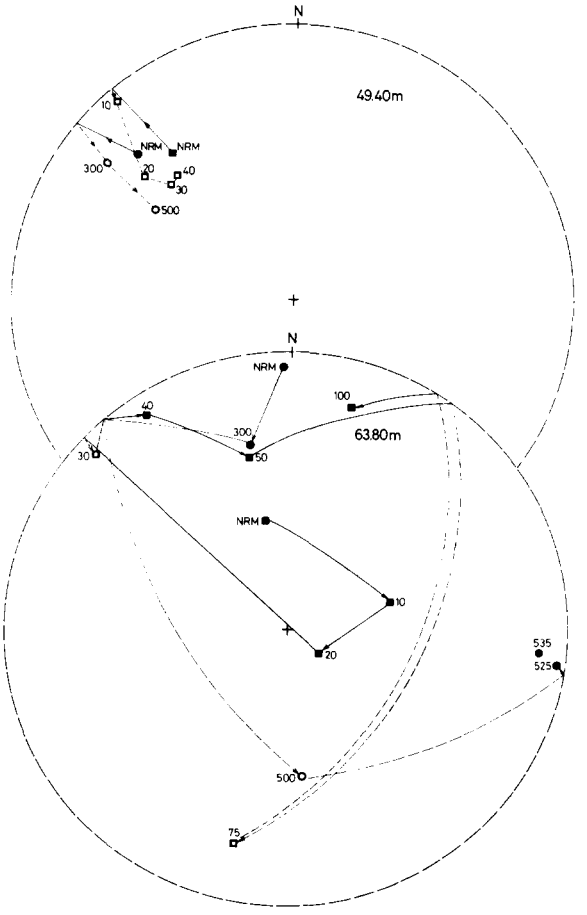


Fig. 5. Stereographic plot of magnetic directions on progressive AF and thermal demagnetization of sister specimens (squares: AF; circles: thermal). The directions are referred to the horizontal plane of the collecting site. Full circles and squares: vectors pointing downward; open circles and squares: vectors pointing upward.

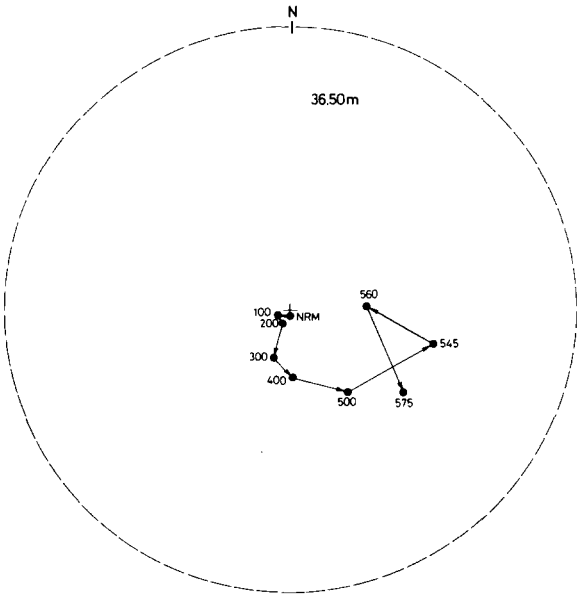


Fig. 6. Directional change in NRM of a pilot specimen with a steady increase in susceptibility above 500°C. The directions are referred to the horizontal plane of the collecting site.

the sister specimens respond to both thermal and AF treatment similarly (Fig. 5.: 49.40 m) but not always (Fig. 5.: 63.80 m). To be on the safe side, thermal treatment was employed for all samples, especially because mineralogical changes on heating were negligible or had no effect on the original NRM. For example, mineralogical change is indicated by the susceptibility curve in Fig. 3, 36.40 m and 94.00 m, but the NRM direction remains basically the same as before the increase (Fig. 6).

From the analysis of the NRM of the pilot specimens it is most probable that the polarity of the remanence surviving heating to 500°C reflects the polarity of the geomagnetic field at the time of deposition. Relatively strongly magnetized samples (mostly from the lower part of the section) were therefore cleaned at two temperatures: 500°C and 545°C, respectively. Weak samples were treated progressively at 300°, 400° and 500°C. Where NRM dropped below noise level below 500°C, NRM cleaned at lower temperature has been used.

A fold test was carried out at two levels. After the correction for local variation of bedding atti-

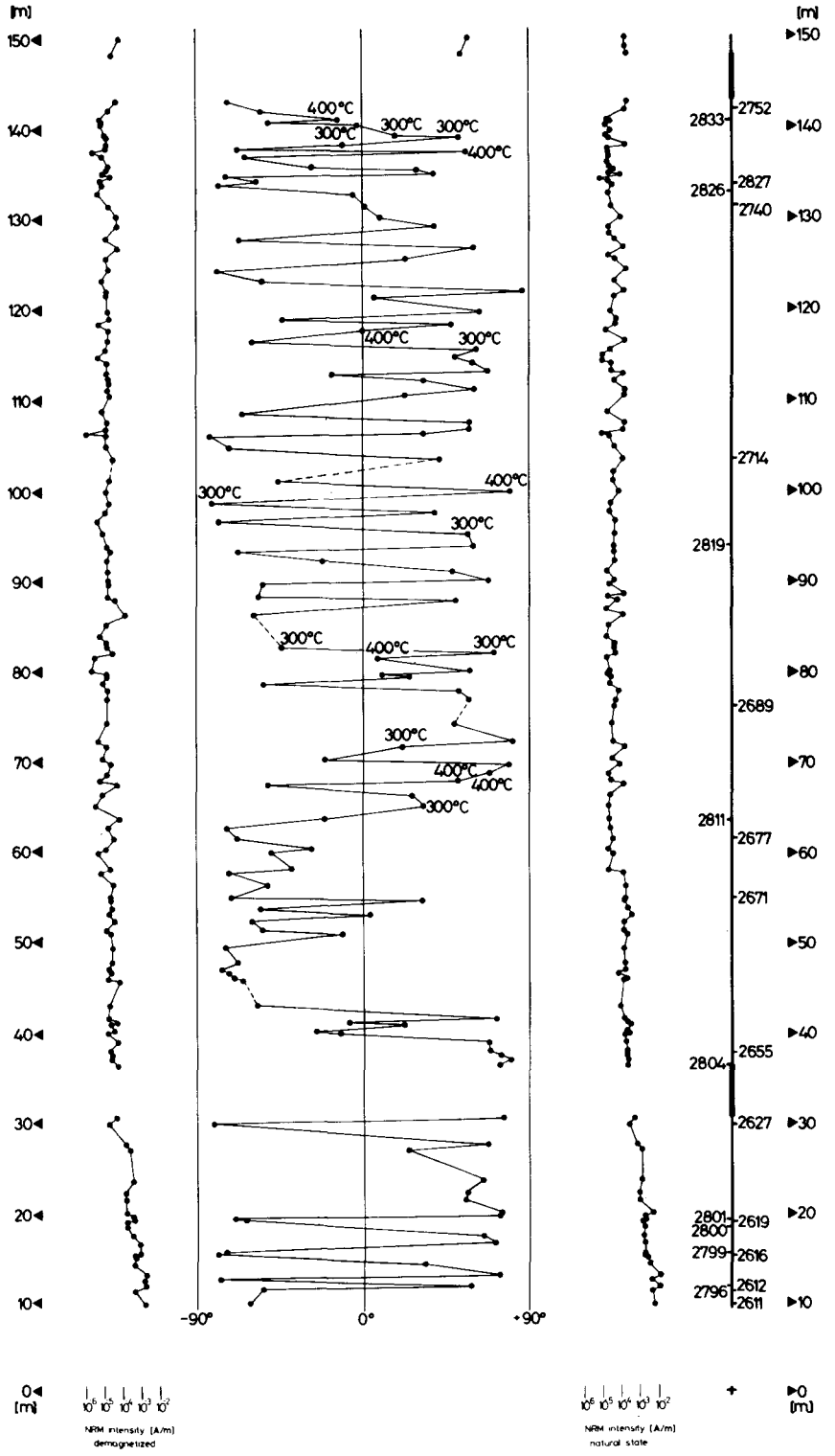


Fig. 7. VGP latitudes for the Sümeg section and the variation of remanence intensity in the natural state and after demagnetization. Arbitrary depths are measured from the base of the Kimmeridgian.

tude within the section the directions become more closely grouped: $\alpha_{95} = 17.4°$ before tilt correction changes to; $\alpha_{95} = 14.0°$ on correction. (For this calculation an equal number of samples with "regular" and "irregular" attitude have been used: $N = 22$). The improvement , however, is not significant on the 95% level [7].

The widely different site mean directions of the Sümeg section (tectonic position 300/105) and Tithonian locality Lókut (tectonic position 350/18) [8] become very similar after correction for tilt. The palaeomagnetic poles after correction are as follows:

Sümeg:
lat. 9°, long. 302°; $\alpha_{95} = 8.5°$; $N = 45$

Lókut:
lat. 12°, long. 307°; $\alpha_{95} = 4.1°$; $N = 14$

For this calculation samples cleaned at lower temperatures than 545°C from the Sümeg section were excluded because their remanence may not always have been sufficiently cleaned, even if the overprint does not effect the polarity. Unfortunately, the statistical significance of the fold test could not be judged since the precisions of the two populations are different. Even so, the closeness of the two palaeopoles after correction, in spite of the great difference in tectonic position, suggests that the magnetization predates the Aptian folding.

4. Magnetic stratigraphy

Virtual palaeomagnetic polar (VGP) latitudes have been calculated, a method widely used in Mesozoic magnetostratigraphy studies. VGP latitude is a useful paramater since the site latitude at the time of magnetization gives a clearly defined magnetic stratigraphy regardless of the pole position and the VGP is a function of both declination and inclination.

The VGP latitudes for the Sümeg section are shown in Fig. 7, after demagnetization at 500–545°C, except for a few points labelled with the temperature of the cleaning. The labelled samples are unusually weak compared to the others. This limited the demagnetization that could be performed, though higher temperature would have been desirable to erase the overprint completely.

Two gaps occur in the coverage (thick line on the depth scale in Fig. 7): between 30 and 40 m and between 140 and 150 m. The upper 70 m of the section gives less tightly grouped directions, probably due to the generally lower NRM intensities.

5. Discussion

The polarity sequence (Fig. 9) has been derived from Fig. 7. The three calibration points for the sequence of magnetic reversals at Sümeg are based on Calpionellids. The Calpionellid zones are referred to the modified London time scale ([2,9], E. Filácz, personal communication).

11.60 m: *Saccocoma* sp., *Stomiosphaera mollucana* Wanner, *Cadosina parvula* Nagy, *Cadosina radiáta* Vogler.

15.40 m: *Saccocoma* sp., *Cadosina parvula* Nagy, *Cadosina malmica* (Borza), *Cadosina lapidosa* Vogler, *Stomiosphaera mollucana* Wanner.

19.20 m: *Saccocoma* sp., *Stomiosphaera mollucana* Wanner, *Cadosina lapidosa* Vogler, *Cadosina parvula* Nagy, *Globochaete alpine* Lombard.

55.00 m: *Calpionella alpina* Lombard, *Calpionella elliptica* Cadisch.

61.60 m: *Calpionella alpina* Lorenz, *Calpionella elliptica* Cadisch, *tintinnopsella carpathica* (Murgenau et Filipescu), *Crassicollaria parvula* Remane, *Nannoconus steimanni* Kamptner.

63.60 m: *Calpionella alpina* Lorenz, *Calpionella elliptica* Cadisch, *Crassicollaria parvula* Remane, *Remainella cadischiana* (colom), *Nannoconus steimanni* Kamptner.

76.20 m: *Calpionella alpina* Lorenz, *Calpionella elliptica* Cadisch, *Crassicollaria parvula* Remane, *Remaniella cadischiana* (colom), *Nannoconus steimanni* Kamptner.

132.60 m: *Calpionella alpina* Lorenz, *Calpionella elliptica* Cadisch, *Tintinnopsella carpathica* (Murgenau et Filipescu), *Nannoconus steimanni* Kamptner, Radiolaria, *Cadosina lapidosa* Vogler, *Cadosina fusca* Wanner.

The age ranges of the diagnostic Calpionellids are plotted against the arbitrary distance from the base of the Kimmeridgian (Fig. 8). The graph suggests a change in sedimentation rate between 60

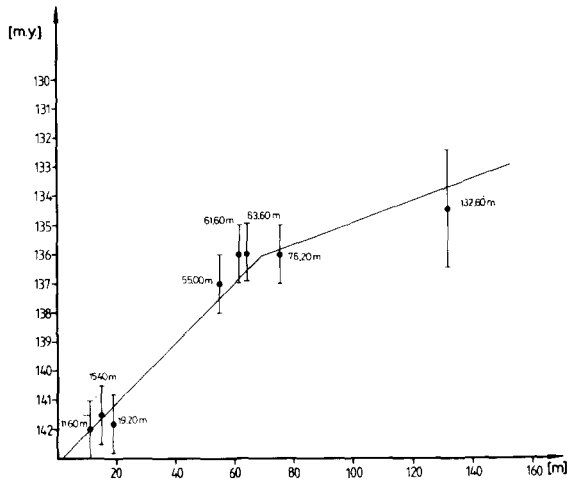


Fig. 8. Graph of the ranges of the diagnostic Calpionellids plotted against the distance from the base of the Kimmeridgian.

and 80 m. For this reason putting the magnetic reversals in a geochronologic framework means stretching the lower and compressing the upper part of the section according to their different sedimentation rate suggested by the palaeontological data: 10 m/Myr up to 70 m, 30 m/Myr from 70 to 150 m.

The mixed polarity interval represented by the Sümeg section is the first on land covering the whole time period from Early Kimmeridgian to Late Berriasian. The only comparison of the polarity sequence is thus with oceanic magnetic anomalies. The dating of the Larson and Hilde [1] time scale (Fig. 9) is approximate, and based on similar procedures to the dating at Sümeg. The difference is that the Sümeg polarity scale is dated at three points instead of two and the interval of interpolation is much narrower than for the oceanic reversal sequence.

Some common trends are visible in both polarity time scales. The most striking common feature is a relatively long reversed zone in the Tithonian (anomaly M17 in the oceanic pattern). In fact it is M17 + M16 and the short normal interval between that correlates quite well with the single long reversed zone at Sümeg, disrupted only by two intermediate directions tending to be normal. (Fig. 7, 53 and 54 m).

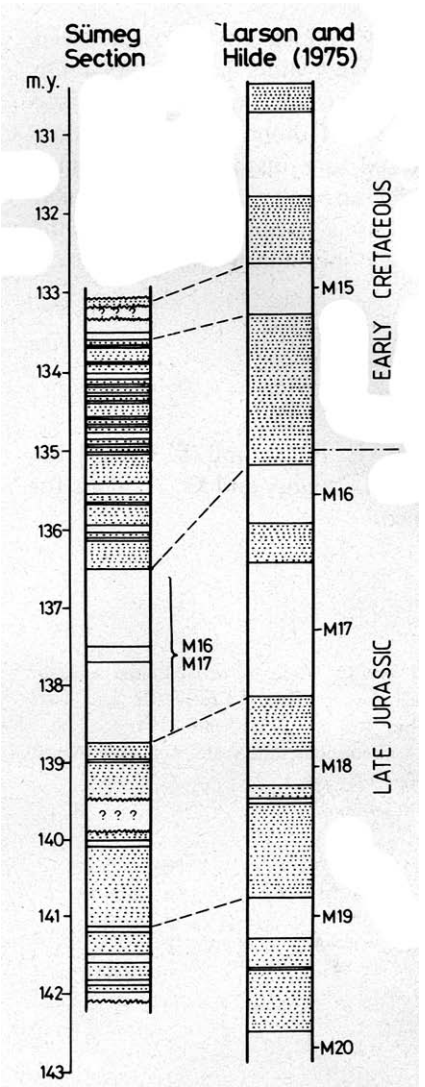


Fig. 9. Comparison of the Sümeg magnetic stratigraphy with that derived from oceanic magnetic anomalies by Larson and Hilde [1].

Both reversal patterns are similar below M17 in the sense that normal polarity is predominant.

Above anomaly M16 there is a long normal zone in the oceanic polarity scale. It seems, however, from the study of the land section at Sümeg, that an apparent quiet interval in the oceanic pattern may have been produced by frequent reversals.

Similar phenomena have been observed for the Jurassic quiet interval [10,11].

The details of the Sümeg polarity scale need confirmation by studying other land sections. On the other hand, the existence of a characteristic reversed zone in the Tithonian has been confirmed. This reversed zone might serve as a starting point for correlating other sections of similar age and perhaps as a means of defining the Jurassic/Cretaceous boundary palaeomagnetically.

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